

PROFINITE GROUPS

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1. INTRODUCTION

TODO

2. PRELIMINARIES

2.1. Limits and Inverse Limits. There are two equivalent definitions for the (category theoretic) limit. One involving cones and the other involving natural transformations. We give the cone definition and leave the other as an exercise.

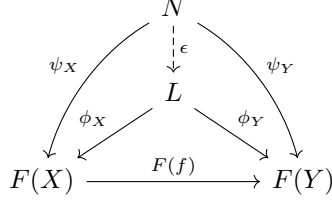
For a functor $F : \mathcal{J} \rightarrow \mathcal{C}$, the image of $\mathcal{J}$ is called a diagram of shape $\mathcal{J}$. Informally, $\mathcal{J}$ can be thought of as an “index category”. A **cone** (M, ϕ) to F is an object M of $\mathcal{C}$ along with a family of morphisms $\phi_X : M \rightarrow F(X)$, for each X in $\mathcal{J}$, such that the following diagram commutes for each $f : X \rightarrow Y$.

$$\begin{array}{ccc}
 & M & \\
 \phi_X \swarrow & & \searrow \phi_Y \\
 F(X) & \xrightarrow{F(f)} & F(Y)
 \end{array}$$

Defining a limit is now easy, a limit is a universal cone. Formally,

Definition 2.1. The **limit** of a functor $F : \mathcal{J} \rightarrow \mathcal{C}$ is a cone (L, ϕ) to F such that any other cone (N, ψ) to F , there is a *unique* morphism $\epsilon : N \rightarrow L$ such that

$\psi_X = \phi_X \circ \epsilon$ for all X in $\mathcal{J}$.



We denote the limit of the functor F by $L = \lim F$.

Intuitively, the limit construction subsumes the entire diagram and all its information into a single object $\lim F$ as far as the morphisms into the diagram are concerned. We illustrate this with an example. Further, it generalises many constructions such as the product, pullbacks, etc.

Consider a poset category on the set $(C, \leq)$. Let $S \subset C$ be any subset. We aim to find $\lim S$ (abusing notation for the inclusion functor). In this category, a cone from M to S is essentially an upper bound of S since M has a morphism to each of the elements of S . Now, $\lim S$ is an object of C such that any upper bound has a unique morphism to it, i.e. it is an upper bound of S that is smaller than all other upper bounds of S . Sound familiar? Indeed, $\lim S = \sup S$.

We leave the alternate definition of the limit as an exercise.

Exercise 2.1. For object M of $\mathcal{C}$, define the constant functor $c_M : \mathcal{J} \rightarrow \mathcal{C}$ where all objects map to M and all morphisms map to id_M . Show that a cone from M to F is a natural transformation from c_M to F .

Exercise 2.2. Show that $L = \lim F$ if there is a natural transformation $\eta : c_L \rightarrow F$ such that any other natural transformation $\epsilon : c_M \rightarrow F$ factors through η .

We are mainly concerned with inverse limits of topological groups, so we first develop the theory of topological inverse limits. The reader should be warned that *inverse limits* are limits and *direct limits* are colimits.

We call a partially ordered set I **directed** if any two elements have a common upper bound. More precisely, for each $x, y \in I$ there is $z \in I$ such that $x, y \leq z$.

An **inverse system** of topological spaces is a contravariant functor $F : \mathcal{I} \rightarrow \mathbf{Top}$ where $\mathcal{I}$ is the poset category on the set I . Expanding this definition out, an inverse system is a collection of topological spaces $\{X_\alpha\}_{\alpha \in I}$ with continuous maps $f_{\alpha\beta} : X_\beta \rightarrow X_\alpha$ whenever $\alpha \leq \beta \in I$.

Definition 2.2. Let I be a directed set and let $(X_\alpha, f_{\alpha\beta})$ be an inverse system of topological spaces with index set I . The inverse limit of the above system is the limit of the contravariant functor $F : \mathcal{I} \rightarrow \mathbf{Top}$ associated to the inverse system.

We denote the inverse limit by $\varprojlim_{\alpha \in I} X_\alpha$ and we shall drop the subscript when there is no risk of confusion. The canonical projections will be denoted by $\pi_\beta : \varprojlim X_\alpha \rightarrow X_\beta$ for each $\beta \in I$.

The question of existence and nonemptiness still remains. We prove below that the inverse limit can be constructed as a subspace of a product. Secondly, inverse limits need not always be nonempty but in the case that we are interested with, the spaces are sufficiently nice and inverse limits are, in fact, nonempty.

Proposition 2.3. *Let I be a directed set and let $(X_\alpha, f_{\alpha\beta})$ be an inverse system of topological spaces with index set I . Then the inverse limit exists and is given by*

$$\varprojlim_{\alpha \in I} X_\alpha = \left\{ x \in \prod_{\alpha \in I} X_\alpha \mid \pi_\alpha(x) = f_{\alpha\beta} \circ \pi_\beta(x) \text{ if } \alpha \leq \beta \in I \right\}.$$

Proof. Exercise. □

Observe that we topologise $\varprojlim X_\alpha$ with the subspace topology of $\prod X_\alpha$ which itself has the product topology. Hence the topology on $\varprojlim X_\alpha$ is the initial topology with respect to the canonical projections.

Proposition 2.4. *Let I be a directed set and let $(X_\alpha, f_{\alpha\beta})$ be an inverse system of nonempty compact Hausdorff spaces with index set I and let $X = \varprojlim X_\alpha$. Then X is compact Hausdorff and nonempty.*

Proof. Clearly X is Hausdorff since the T_2 -axiom is closed under product and subspaces. By Tychonoff's Theorem we get that $\prod X_\alpha$ is compact. To show X is compact, it suffices to show that X is a closed subspace.

For each $\alpha \leq \beta \in I$, let $\Gamma_{\alpha\beta} = \{x \in \prod X_\alpha \mid \pi_\alpha(x) = f_{\alpha\beta} \circ \pi_\beta(x)\}$. Since X_α is Hausdorff, the diagonal set $D = \{(s, s) \in X_\alpha \times X_\alpha \mid s \in X_\alpha\}$ is closed. Moreover, $\Gamma_{\alpha\beta} = (f_{\alpha\beta} \times \text{id}_{X_\alpha})^{-1}(D)$, hence it is closed. Furthermore,

$$(1) \quad X = \bigcap_{\alpha \leq \beta \in I} \Gamma_{\alpha\beta}$$

which proves our first assertion.

We now prove that X is nonempty. Let $n \in \mathbb{N}$ and let $\alpha_i \leq \beta_i \in I$ for $i = 1, \dots, n$. We show that

$$\Gamma_{\alpha_1\beta_1} \cap \dots \cap \Gamma_{\alpha_n\beta_n} \neq \emptyset.$$

We construct an element $(x_\alpha)_{\alpha \in I}$ that is in the aforegiven set. Since I is directed, there is $\gamma \in I$ such that $\beta_i \leq \gamma$ for $i = 1, \dots, n$. Pick any $x_0 \in X_\gamma$ and put $x_\nu = f_{\nu\gamma}(x_0)$ for $\nu = \alpha_1, \beta_1, \dots, \alpha_n, \beta_n$. Pick the rest of the $x_\alpha \in X_\alpha$ arbitrarily. Our assertion follows from [Equation 1](#) and the finite intersection property. □

2.2. Topological Groups. A topological group G is a topological space with a compatible group structure. That is, the group operation $G \times G \rightarrow G$ sending $x \times y$ to $x \cdot y$ and inversion $G \rightarrow G$ sending x to x^{-1} are continuous maps. For the general theory, we refer the reader text such as [\[Bou98\]](#) and [\[Mun99\]](#).

Let G be a topological group throughout. We state a few important results.

Exercise 2.3. The component C of the identity element $e \in G$. Then C is a closed normal subgroup of G . Further, G/C is totally disconnected.

Exercise 2.4. Let G be compact and let $H \leq G$. Then H is open iff H has finite index and is closed.

TODO

2.3. Totally Separated Spaces. Recall that quasicomponents are defined by the equivalence relation “ $x \sim y$ if there is no separation (U, V) of X with $x \in U$ and $y \in V$ ”. It can be shown that QC_x is the intersection of all clopen neighbourhoods of x . Thus QC_x is closed. It can also be shown that $C_x \subset QC_x$ for all $x \in X$.

Recall that totally disconnected spaces are where all connected components are singletons and totally separated spaces are where all quasicomponents are singletons.

Theorem 2.5. *If X is compact Hausdorff then quasicomponents agree with the connected components. In particular, if X is compact Hausdorff then totally disconnected $\iff$ totally separated.*

Proof. It is clear that $C_x \subset QC_x$, it is enough to show that QC_x is connected. Suppose not, then there is a separation (X_1, X_2) of QC_x with $x \in X_1$. Observe that X_1 is closed in X since $X_1 = G \cap QC_x$ where G is closed in X , similarly for X_2 . By the normality of X , there are open $U, V \subset X$ such that $X_1 \subset U$ and $X_2 \subset V$.

Let $\{S_\alpha\}_{\alpha \in I}$ denote the set of all clopen neighbourhoods of x . Recall that $QC_x = \bigcap_{\alpha \in I} S_\alpha$, we claim that some $S_\alpha \subset U \cup V$. First observe that $S_\alpha \cap (X \setminus U \cup V)$ is compact. Now,

$$\bigcap_{\alpha \in I} (S_\alpha \cap (X \setminus U \cup V)) = QC_x \cap (X \setminus U \cup V) = \emptyset.$$

By the finite intersection property of compact sets, there are $\{\alpha_i\}_{i=1}^n$ such that $\bigcap_{i=1}^n (S_{\alpha_i} \cap (X \setminus U \cup V)) = \emptyset$. Define $S = \bigcap_{i=1}^n S_{\alpha_i}$, then $S \subset U \cup V$ and S is clopen. We now show that $S \cap U$ is clopen, indeed

$$\overline{S \cap U} \subset S \cap \overline{U} = S \cap (U \cup V) \cap \overline{U} = S \cap U$$

since $V \cap \overline{U} = \emptyset$. Since $S \cap U$ is a clopen neighbourhood of x , we have $QC_x \subset S \cap U$ which forces $X_2 = \emptyset$, a contradiction. $\square$

3. PROFINITE GROUPS

Lemma 3.1. *For a Hausdorff space T , the following are equivalent:*

- (i) T is homeomorphic to the topological inverse limit of finite discrete spaces,
- (ii) T is compact and has a clopen neighbourhood basis at each point,
- (iii) T is compact and totally separated.

Proof. We first show (i) $\Rightarrow$ (ii). It follows from Theorem 2.4 that T is compact. It is given that $T = \varprojlim V_\alpha$ where V_α are finite discrete spaces. Then $\pi_\alpha^{-1}(x)$ for all $x \in V_\alpha$ form a basis for T . These are clopen since V_α have the discrete topology.

Now we show (ii) $\Rightarrow$ (iii). Let $x, y \in T$ be distinct. Since T is Hausdorff, there are open $U \ni x$ and $V \ni y$. By (ii) there is a clopen neighbourhood $x \in S \subset U$. Then $(S, X \setminus S)$ is a separation of X such that $x \in S$ and $y \in X \setminus S$, so the quasicomponents of X must be singletons.

Finally we show (iii) $\Rightarrow$ (i). Let I be the set of all equivalence relations $R \subset T \times T$ on T such that T/R is finite and discrete. This is nonempty since $T \times T \in I$. Order I by reverse inclusion. To show that I is directed, we show that since $R_1 \cap R_2 \in I$. Indeed, $R_1 \cap R_2$ is an equivalence relation and the map $\phi : T/R_1 \cap R_2 \rightarrow T/R_1 \times T/R_2$ given by $\phi = \pi_1 \times \pi_2$ is injective which gives us that $T/R_1 \cap R_2$ is finite and discrete since $T/R_1 \times T/R_2$ is finite and discrete. We claim that the canonical map $\phi : T \rightarrow \varprojlim_{R \in I} T/R$ is a homeomorphism.

First we prove that ϕ is surjective. Let $\{t_R\}_{R \in I} \in \varprojlim_{R \in I} T/R$. Then $\pi_R^{-1}(t_R)$ is nonempty and closed, hence compact. Since I is directed and using a similar argument as in [Theorem 2.4](#), finite intersections of these sets is nonempty. Our claim follows from the finite intersection property and the identity

$$\phi^{-1}(\{t_R\}_{R \in I}) = \bigcap_{R \in I} \pi_R^{-1}(t_R).$$

Now we show that ϕ is injective. Let $x, y \in T$ distinct. Since T is totally separated, there is a clopen $U \ni x$ with $y \notin U$. Let R be the equivalence relation related to the partition U and $X \setminus U$. Then T/R is finite and discrete. Moreover, $\phi(x)$ and $\phi(y)$ will differ in the R -th slot.

Since a continuous bijection between compact Hausdorff spaces is a homeomorphism, our claim follows. $\square$

These equivalences define a **Stone space** or **Profinite space**.

Proposition 3.2. *For a Hausdorff topological group G the following are equivalent:*

- (1) G is a topological inverse limit of finite discrete groups,
- (2) G is compact and e has a neighbourhood basis of clopen normal subgroups,
- (3) G is compact and totally separated.

Proof. (i) $\Rightarrow$ (iii). It follows from [Theorem 2.4](#) that G is compact. Moreover, the product and subspace of totally disconnected spaces is totally disconnected, so G is totally disconnected. $\square$

3.1. Examples.

4. G-MODULES

5. COHOMOLOGY OF PROFINITE GROUPS

REFERENCES

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